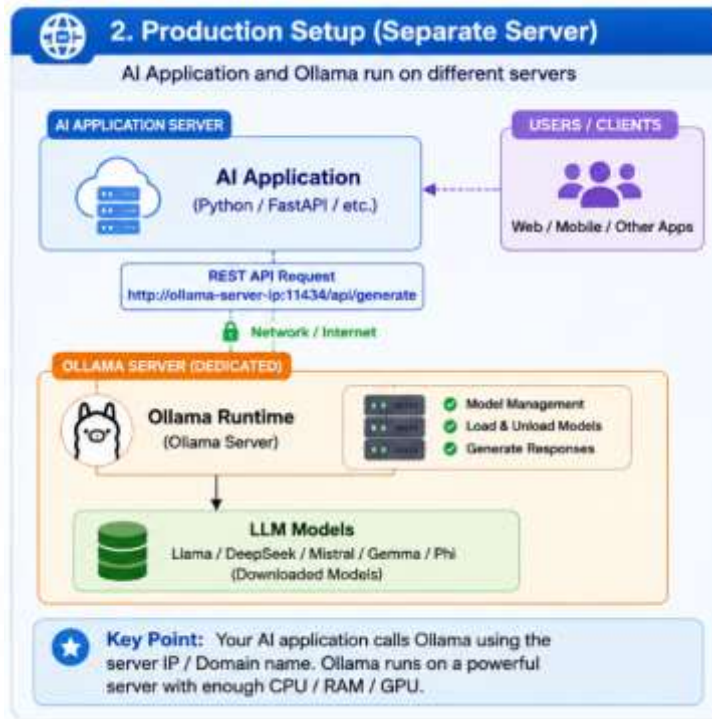


Production-Ready Local LLM Deployments

- So far, we have installed Ollama and Open-Weight LLMs on our personal computers and executed some requests. This setup is ideal for learning, development, and testing.
- Running Local LLMs on personal computers may result in higher latency and lower throughput due to limited hardware resources.
- Local LLMs running on a personal computer are typically available only to applications on that machine.
- In production, organizations deploy Local LLMs on dedicated Linux servers or Kubernetes platforms to improve performance, scalability, and availability.
- **Typical Production Infrastructure**
- Organizations typically deploy Ollama on:
 - ✓ High-performance Linux servers
 - ✓ Servers with 64–128 GB (or more) RAM
 - ✓ Multi-core CPUs
 - ✓ NVIDIA GPUs for accelerated inference
 - ✓ High-speed SSD storage
 - ✓ Docker containers
 - ✓ Kubernetes platforms (OKE, EKS, AKS, GKE)
 - ✓ Multiple Ollama instances behind a Load Balancer
- **Note:** Local LLMs are often preferred when organizations want greater control over their data, infrastructure, and security. Although modern AI providers and cloud platforms offer robust security features and compliance options, some organizations still choose Local LLMs to keep sensitive data entirely within their own environment.
- **Note:** For better performance during practice, you can deploy Ollama on a standalone cloud instance with higher CPU, RAM, and GPU resources. However, this approach involves additional cloud costs.

Production-Ready Local LLM Deployments



Production-Ready Local LLM Deployments

- So far, we have installed Ollama and Open-Weight LLMs on our personal computers and built AI applications using them. This setup is ideal for learning, development, and testing.
- However, most personal computers have limited hardware resources, such as CPU cores, RAM, and often no dedicated GPU. As a result, running Local LLMs on a laptop may lead to higher response latency and lower throughput, especially when using larger models.
- Another limitation is that the Local LLM is available only on your own computer. Other applications or users in your organization cannot access it unless you expose it over a network.
- In production environments, organizations typically deploy Ollama on dedicated Linux servers or on Kubernetes platforms such as OKE, EKS, AKS, or GKE. These environments generally provide high-core-count CPUs, large amounts of RAM (for example, 128 GB or more), NVIDIA GPUs, high-speed SSD storage, and multiple Ollama instances behind a Load Balancer to improve performance, scalability, and availability.
- Note: Local LLMs are often preferred when organizations want greater control over their data, infrastructure, and security. Although modern AI providers and cloud platforms offer robust security features and compliance options, some organizations still choose Local LLMs to keep sensitive data entirely within their own environment.